

SwaVid: Product Engineering Take-Home Challenge

1. What This Challenge Is About

This is an open-ended product and engineering challenge based on an imperfect real-world situation.

There is no predetermined solution, hidden test case, or expected technology stack. We want to understand how you:

- Make sense of an ambiguous problem
- Identify what matters most
- Decide what should and should not be solved
- Design a practical experience for the people involved
- Build a small but functional version of your idea
- Use AI tools thoughtfully

AI tools are allowed and expected. You may use ChatGPT, Claude, Cursor, Copilot, v0, Lovable, Replit, or any other tool you normally work with.

We are evaluating your decisions and judgment, not whether you manually wrote every line of code.

2. Time and Scope

You will have 5–7 days to complete the challenge so that you can fit it around your existing schedule. We do not expect you to spend the entire week working on it.

Please limit your total effort to approximately **5–6 hours**.

A reasonable division of time might be:

Phase	Suggested time
Understand the scenario	0.5-1 hour
Decide your approach	1-2 hour
Build a working version	3-5 hours
Prepare the submission	2-3 hour

If you cross 10 hours, stop building and submit what you have. Explain honestly what remains incomplete.

Choosing an appropriate scope is part of the challenge.

3. Pick One Scenario

Choose one of the following four scenarios. All four are equally valid and will be evaluated using the same criteria.

Scenario A — The School Bus Network

A private school operates 18 buses carrying approximately 750 students every morning and afternoon.

Routes were originally created when the school had fewer than 400 students. Since then, new residential areas have developed, students have changed addresses, and additional stops have been added wherever parents requested them.

The transport manager explains:

“Some children spend almost two hours on the bus even though they live only nine kilometres from school. Two buses regularly arrive half empty, while three others are overcrowded. Parents call whenever a bus is late, but I often do not know where the delay happened.

Every few months, we manually move stops between routes. Fixing one route usually creates a problem somewhere else. If one driver is absent or a bus breaks down, the entire morning becomes unpredictable.”

The school has the following information:

- Student names, classes and home addresses
- Assigned bus and stop for each student
- Planned route sequences
- Approximate pickup and drop times
- Monthly fuel expenses for each bus
- Driver attendance records
- Informal records of repeated delays and parent complaints

However, the address data is inconsistent. Some entries contain exact addresses, some contain only housing-society names, and others are outdated.

Additional constraints:

- Younger students should not be the first to board or the last to be dropped
- Certain roads cannot accommodate large buses
- Route changes can create strong objections from parents
- Drivers are not comfortable entering information while driving

- The school cannot replace its current bus fleet this year
- Any major route change must be communicated before the beginning of the next term
- The transport manager primarily works from an Android phone

The next school term begins in five weeks.

Scenario B — The Diagnostic Laboratory

A neighbourhood diagnostic laboratory processes blood tests, urine tests and routine health packages for approximately 180 patients each day.

Patients arrive through three channels:

- Walk-ins
- Home sample collections
- Referrals from nearby doctors and clinics

The laboratory owner explains:

“Patients become anxious when reports are delayed, but we often cannot tell them where the delay is. Sometimes the sample has not reached the lab. Sometimes a machine is overloaded. Sometimes a technician needs the test to be repeated.

The front desk gives patients an estimated delivery time, but it is usually based on experience rather than what is actually happening inside the lab.”

A normal sample passes through several steps:

1. Registration
2. Sample collection
3. Labelling
4. Transportation, if collected from home
5. Testing
6. Technician verification
7. Doctor review for certain reports
8. Report release

Different tests have different processing times. Some tests are performed continuously, while others are processed together only once or twice a day.

Additional constraints:

- A single sample may contain multiple tests with different completion times
- Some abnormal results require manual verification
- Samples can become unusable if stored or transported incorrectly
- Patients may not understand technical laboratory language

- The laboratory handles sensitive medical information
- Home-collection staff use low-cost Android devices with inconsistent internet access
- Technicians wear gloves and cannot repeatedly interact with a screen
- The laboratory already uses billing software, but its internal workflow is mostly tracked through printed labels and verbal updates
- Incorrectly releasing a report is significantly worse than releasing it late
- The owner cannot pause daily operations to introduce a complicated new system

The laboratory wants to improve operations before opening a second location in three months.

Scenario C — The Second-Hand Electronics Market

A local business buys and sells used smartphones, laptops and tablets.

It purchases devices directly from individuals, refurbishes some of them and resells them through a physical store and several online marketplaces.

The owner explains:

“Two phones can look identical but have completely different histories. One may have a replaced screen, one may have water damage, one may be stolen, and one may work perfectly.

My staff checks each device, but every employee checks it differently. Customers ask why one phone costs ₹18,000 and another of the same model costs ₹23,000. I usually cannot show them a clear answer.

We also discover problems after buying devices. By then, the seller is gone and we carry the loss.”

The business handles approximately:

- 25–40 incoming devices each day
- 200–250 devices in current inventory
- 10–15 customer returns per week
- Four employees who inspect and price devices
- Two technicians who perform repairs

The business currently stores information across handwritten intake forms, device photographs, marketplace listings, repair invoices and a shared spreadsheet.

Additional constraints:

- Device serial numbers and identification numbers must be handled carefully
- Sellers sometimes provide incomplete or misleading information
- Cosmetic damage is subjective
- Battery health and performance can change after inspection

- Some faults appear only after extended use
- Repairing a device may increase its selling price but delay the sale
- Marketplace prices change frequently
- Customers expect a warranty, but the business cannot guarantee every component
- Employees have different levels of technical knowledge
- The owner does not want the inspection process to take more than 15 minutes per device
- A wrong purchase decision can eliminate the profit from several successful sales

The owner wants the business to grow without personally inspecting every device.

Scenario D — The District Water-Tanker System

A rapidly growing town supplies piped water to most neighbourhoods, but shortages occur during summer and in areas with weak infrastructure.

During shortages, the municipal office sends water tankers to housing societies, public schools, hospitals and informal settlements.

The officer responsible for tanker distribution explains:

“Every morning, we receive more requests than we can fulfil. Some are genuine emergencies. Some areas receive tankers repeatedly because they have influential representatives. Other areas wait for days.

Residents do not know why one request was accepted and another was rejected. Drivers sometimes reach a location and discover there is no suitable place to unload the water. We also receive complaints that tankers never arrived, while the driver says the delivery was completed.”

The municipality has:

- 14 municipal tankers
- 11 contracted private tankers
- Six filling stations
- Approximately 80–150 requests on a normal summer day
- More than 250 requests during severe shortages
- A small dispatch team working in two shifts

Requests currently arrive through phone calls, written letters, visits to the municipal office and referrals from elected representatives.

Additional constraints:

- Hospitals and schools may require priority treatment
- Informal settlements often lack formal addresses
- Some neighbourhoods have private borewells but still request municipal water
- Tanker capacities differ

- Filling stations operate at different speeds and hours
- Traffic conditions strongly affect delivery time
- Drivers may have basic phones rather than smartphones
- Mobile connectivity is unreliable in some areas
- Water availability at filling stations changes during the day
- Any prioritisation process may be challenged by residents or local representatives
- Publicly revealing household-level requests may create privacy and safety concerns
- The municipality has limited authority over contracted tanker drivers
- The system must continue functioning during power or internet outages

Peak summer begins in six weeks.

4. Your Task

Choose one scenario and create something that materially improves the situation.

The scenario may contain several visible problems. You are not expected to solve all of them.

You must decide:

- What you believe the most important problem is
- Who you are solving it for
- What assumptions you are making
- What you will build within the available time
- What you will deliberately leave out

We are not prescribing the format, interface, technology, or type of solution.

A narrow, coherent and functional submission is better than a large collection of incomplete features.

5. Deliverables

Submit one folder or repository containing the following:

1. README

Keep the README to a maximum of one page. It should explain:

- Which scenario you selected
- The problem you chose to address, in your own words
- Who the primary user is
- What you chose not to solve and why
- The assumptions you made
- How your solution works from beginning to end
- How to run or view it
- What you would do with five additional hours

Include one simple sketch, diagram or flow showing how the solution works.

2. Working Demonstration

Submit at least one of the following:

- A deployed link
- A runnable repository
- A prototype with functional interactions
- A three-to-five-minute screen recording

A screen recording is acceptable when deployment would consume a disproportionate amount of time.

The demonstration should show the main experience working from beginning to end.

6. Evaluation Criteria

We will evaluate the submission across the following dimensions:

Dimension	What we will assess
Problem understanding	Did you identify a meaningful underlying problem rather than reacting only to the most visible symptom?
Prioritisation	Did you make deliberate choices about what to solve first?
Scope discipline	Is the submission appropriately sized for the available time?
User understanding	Does the solution reflect the actual environment, abilities and constraints of its users?
Practical adoption	Could the solution realistically fit into the existing workflow?
Product judgment	Do the decisions form a coherent and useful experience?
Working execution	Does the core flow function well enough to demonstrate the idea?
AI judgment	Did you critically review, correct and direct AI-generated work?
Communication	Can another person clearly understand the problem, decisions and solution?

We are not evaluating:

- Production-grade architecture
- Extensive test coverage
- Pixel-perfect visual design
- A large number of features
- The use of a particular framework

- Whether the solution is ready for commercial deployment

7. Ground Rules

- Complete the challenge independently.
- AI tools, libraries, templates, boilerplates and UI kits are allowed.
- Do not ask another person to design or build the submission for you.
- Do not spend money on software, APIs, hosting or infrastructure.
- Free tiers and mocked data are acceptable.
- Clearly identify any mocked, assumed or manually created data.
- Do not collect or use real sensitive personal information.
- You may simplify the scenario, but document the simplification.
- Incomplete but transparent work is better than polished work you cannot explain.
- Be prepared to discuss every major product and technical decision in your submission.

8. What Happens Next

We will review your submission within 3 Working days.

After that you will be invited to a 45-minute conversation where you will walk us through:

- How you interpreted the scenario
- Which problem you selected
- What alternatives you considered
- What you chose not to build
- What changed while you were working
- Where AI helped
- Where AI produced weak or incorrect output
- What you would improve next

This will be a discussion about your work, not a separate technical test.

Good Luck and Enjoy the process!!